S11MD5T/S21MD3TV/ S21MD4TV

High Noise-resistance Type Phototriac Coupler

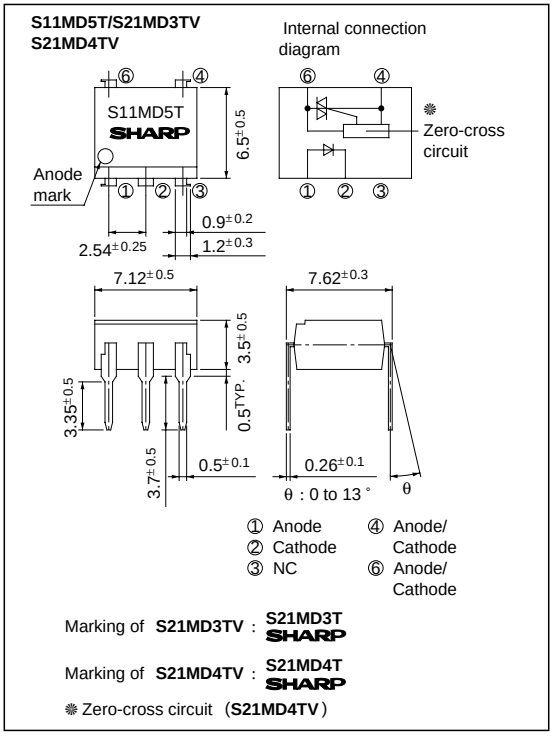
■ Features

1. NO.5 pin completely sealed in the mold for external noise resistance
2. Built-in zero-cross circuit **(S21MD4TV)**
3. High repetitive peak OFF-state voltage.
   - **S11MD5T** V_{DRM} : MIN. 400V
   - **S21MD3TV/S21MD4TV** V_{DRM} : MIN. 600V
4. Isolation voltage between input and output (Viso : 5 000 Vrms)
5. Recognized by UL : recognized, file No. E64380

■ Applications

1. For triggering of power triac

■ Outline Dimensions (Unit : mm)



■ Model Line-ups

100V	**S11MD5T**
200V	**S21MD3TV/S21MD4TV**

■ Absolute Maximum Ratings (Ta = 25°C)

Parameter		Symbol	Rating		Unit
			S11MD5T	S21MD3TV/S21MD4TV	
Input	Forward current	I_F	50		mA
	Reverse voltage	V_R	6		V
Output	RMS ON-state current	I_T	0.1		A_{rms}
	*1Peak one cycle surge current	I_{surge}	1.2		A
	Repetitive peak OFF-state voltage	V_{DRM}	400	600	V
*2Isolation voltage		V_{iso}	5 000		V_{rms}
Operating temperature		T_{opr}	- 30 to + 100		°C
Storage temperature		T_{stg}	- 55 to + 125		°C
*3Soldering temperature		T_{sol}	260		°C

*1 Sine wave
*2 40 to 60% RH, AC for 1 minute, f = 60Hz
*3 For 10 seconds

" In the absence of confirmation by device specification sheets, SHARP takes no responsibility for any defects that occur in equipment using any of SHARP's devices, shown in catalogs, data books, etc. Contact SHARP in order to obtain the latest version of the device specification sheets before using any SHARP's device."

■ Electro-optical Characteristics

(Ta= 25˚C)

Parameter			Symbol	Conditions	MIN.	TYP.	MAX.	Unit
Input	Forward voltage	**S11MD5T/S21MD4TV**	V_F	I_F = 20mA	-	1.2	1.4	V
		S21MD3TV		I_F = 30mA				
	Reverse current		I_R	V_R = 3V	-	-	10^{-5}	A
Output	Repetitive peak OFF-state current		I_{DRM}	V_{DRM} = Rated	-	-	10^{-6}	A
	ON-state voltage	**S11MD5T**	V_T	I_T = 0.1A	-	1.3	2.0	V
		S21MD3TV/S21MD4TV			-	1.7	2.5	V
	Holding current		I_H	V_D = 6V	0.1	1	3.5	mA
	Critical rate of rise of OFF-state voltage	**S11MD5T/S21MD4TV**	dV/dt	V_{DRM} = 1/√2 Rated	100	-	-	V/µs
		S21MD3TV			500	-	-	V/µs
	Zero-cross voltage	**S21MD4TV**	V_{OX}	Resistance load I_F = 15mA	-	-	35	V
Transfer characteristics	Minimun trigger current		I_{FT}	V_D = 6V R_L = 100Ω	-	-	10	mA
	Isolation resistance		R_{ISO}	DC500V 40 to 60% RH	5 x 10^{10}	10^{11}	-	Ω
	Turn-on time	**S11MD5T**	t_{on}	V_D = 6V, I_F = 20mA*4 R_L = 100Ω	-	80	200	µs
		S21MD3TV			-	-	100	µs
		S21MD4TV			-	20	50	µs

*4 **S21MD3TV** : I_F=30mA

Fig. 1 RMS ON-state Current vs. Ambient Temperature

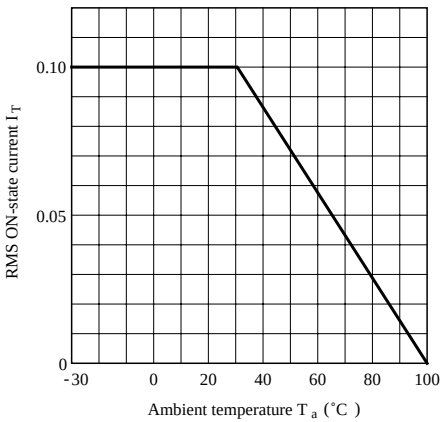


Fig. 2 Forward Current vs. Ambient Temperature

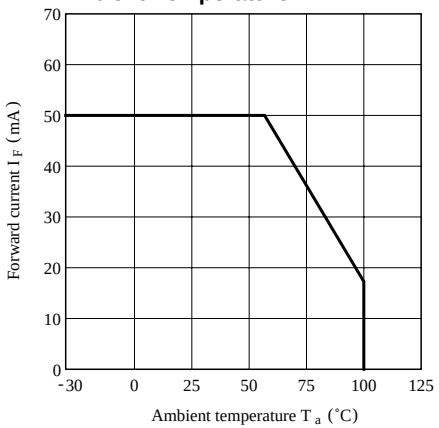


Fig. 3 Forward Current vs. Forward Voltage

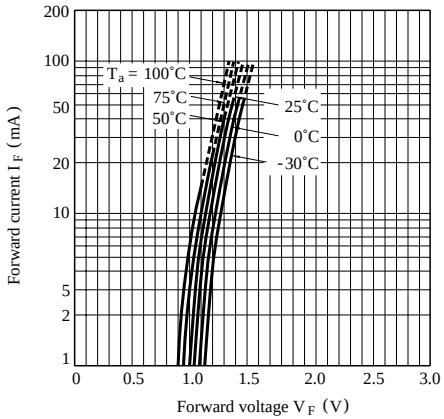


Fig. 4 Minimum Trigger Current vs. Ambient Temperature

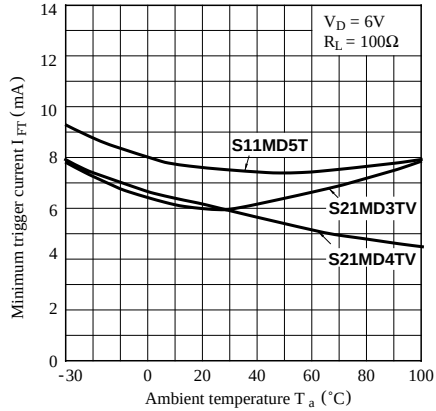


Fig. 5 Relative Repetitive Peak OFF-state Voltage vs. Ambient Temperature

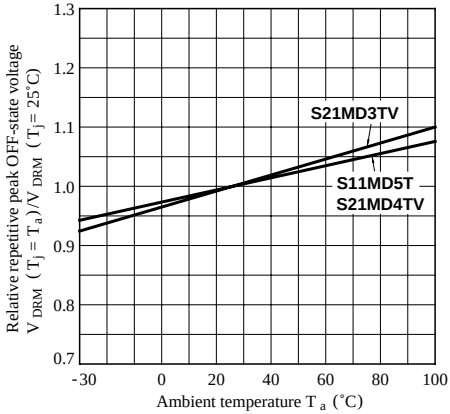


Fig. 6-a ON-state Voltage vs. Ambient Temperature (S11MD5T)

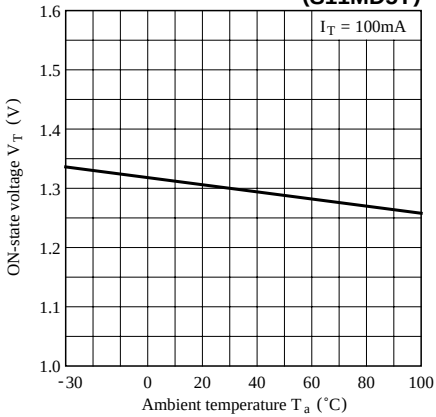


Fig. 6-b ON-state Voltage vs. Ambient Temperature (S21MD3TV/S21MD4TV)

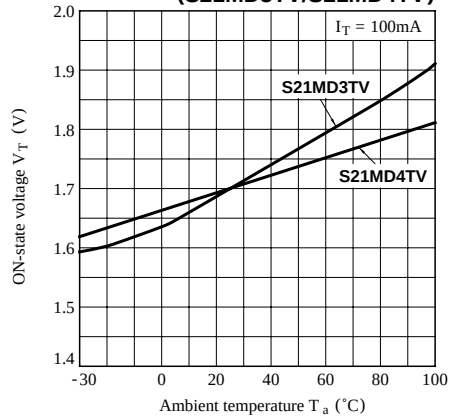


Fig. 7 Holding Current vs. Ambient Temperature

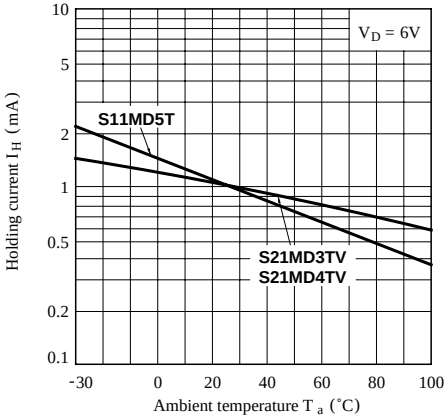


Fig. 8-a Repetitive Peak OFF-state Current vs. OFF-state Voltage (S11MD5T)

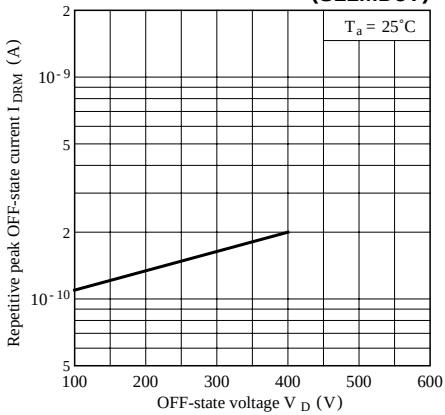


Fig. 8-b Repetitive Peak OFF-state Current vs. OFF-state Voltate (S21MD3TV/S21MD4TV)

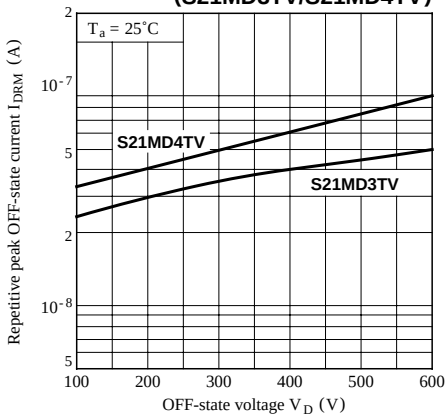


Fig. 9-a Repetitive Peak OFF-state Current vs. Ambient Temperature (S11MD5T)

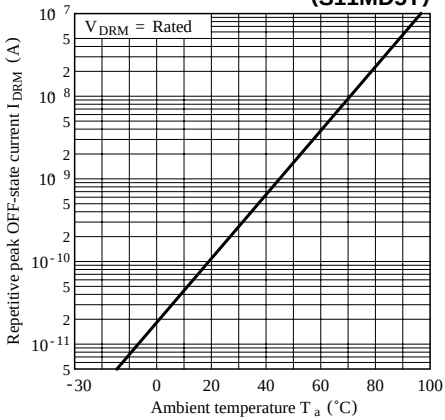


Fig. 9-b Repetitive Peak OFF-state Current vs. Ambient Temperature (S21MD3TV/S21MD4TV)

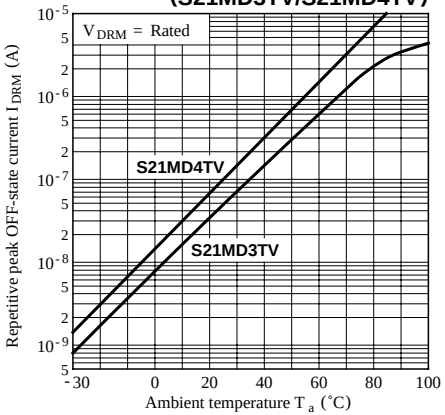


Fig.10 Turn-on Time vs. Forward Current (S11MD5T/S21MD3TV)

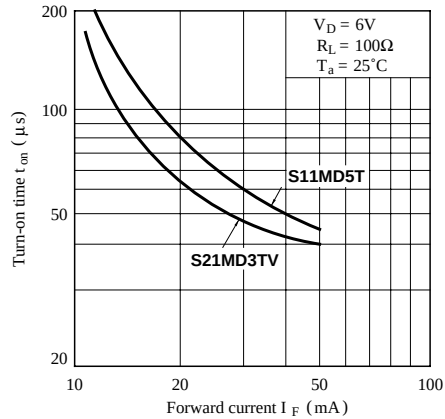


Fig.11 Zero-cross Voltage vs. Ambient Temperature (S21MD4TV)

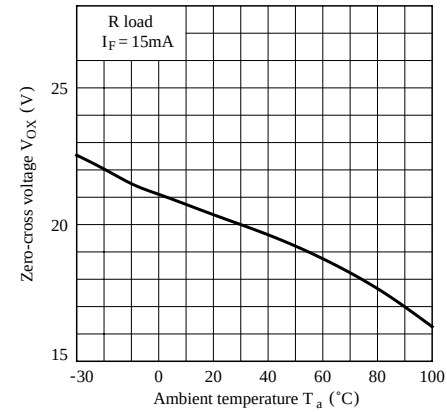
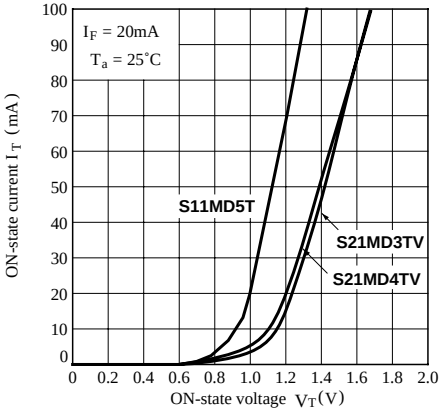
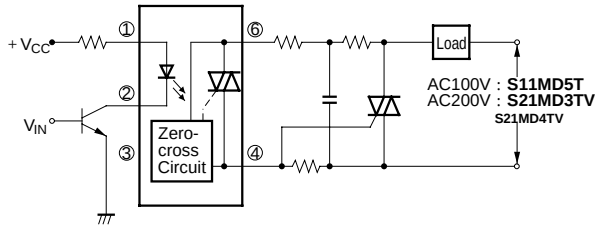


Fig.12 ON-state Current vs. ON-state Voltage



■ Basic Operation Circuit

Medium/High Power Triac Drive Circuit



Note) Please use on condition of the triac for power triggers.
Zero-cross circuit is applied to **S21MD4TV**.

- Please refer to the chapter "Precautions for Use."